

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr. Tamilselvi

QUESTIONS: 1 (1 X 100 = 100)

Total Marks: 100 Duration: 3hrs

10) Develop a functional split model between centralized and distributed radio units.

Code of Conduct:

I M.GOKULAVASAN(192312477) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

1. Understanding of Design

Design Objective: To develop a flexible and efficient functional split architecture that distributes wireless communication processing between a Centralized Unit (CU) and Distributed Radio Units (DU/RU). The design aims to achieve low latency, reduced fronthaul requirements, efficient resource utilization, and scalable operation for 5G/6G wireless networks.

Constraints & Context: Modern wireless networks generate high data rates and require rapid processing for applications such as enhanced mobile broadband, massive IoT, autonomous systems, and real-time communication. Fully centralized architectures require high-capacity fronthaul links, while fully distributed architectures increase processing requirements at individual radio sites. The proposed functional split addresses this trade-off by assigning computationally intensive and coordination-oriented functions to the CU while performing latency-sensitive radio functions closer to the users.

2. Evaluation of Design

Strengths:

- **Fronthaul Optimization:**

By performing selected physical-layer operations at the distributed radio units, the amount of intermediate data transported between the CU and RU is reduced, lowering the required fronthaul capacity.

- **Low Latency:**

Time-critical functions such as modulation, FFT/IFFT, beamforming, and RF processing are performed closer to the antenna, reducing processing and transport delays.

- **Centralized Coordination:**

Network control, scheduling, and resource management are performed centrally, allowing efficient coordination among multiple radio units and cells.

- **Scalability:**

Multiple distributed radio units can be connected to a common centralized unit, allowing the architecture to support dense 5G/6G deployments.

- **Resource Efficiency:**

Computational tasks are divided according to their processing and latency requirements, avoiding unnecessary concentration of all processing at a single location.

Weaknesses & Design Gaps:

- **Fronthaul Dependency:**

The functional split still requires a reliable and sufficiently capable fronthaul connection between the CU and distributed radio units.

- **Synchronization Requirement:**

Distributed radio units must maintain accurate timing and synchronization for coordinated wireless transmission.

- **Processing Distribution Complexity:**

Selecting an inappropriate split may overload either the centralized unit or the distributed units, reducing overall network efficiency.

3. Justification

The proposed architecture uses a hybrid centralized–distributed processing model. Higher-layer processing, scheduling, network control, and resource management are assigned to the Centralized Unit, enabling coordinated resource allocation across multiple radio units.

At the distributed side, baseband processing, modulation, FFT/IFFT, beamforming, and RF operations are performed closer to the antenna. This minimizes latency for time-sensitive operations and reduces the amount of data that must cross the fronthaul interface.

For the proposed MATLAB model, 60% of the processing is assumed to be performed locally at the distributed radio units, while the remaining 40% is handled through the centralized processing path. With multiple radio units generating different traffic loads, the simulation evaluates processing distribution and fronthaul requirements.

This design therefore provides a practical compromise between centralized network coordination and distributed low-latency processing, making it suitable for efficient 5G/6G wireless transmission systems.

4. Suggestions for Improvement

- Adaptive Functional Split:

Implement an adaptive mechanism that dynamically changes the functional split according to traffic load, fronthaul capacity, processing availability, and latency requirements.

- AI-Based Resource Allocation:

Machine-learning techniques can be introduced at the CU to predict traffic demand and dynamically allocate computational resources among distributed radio units.

- Edge-Assisted Processing:

Additional edge computing resources can be deployed near high-traffic radio units to handle processing bursts and prevent overload at the CU.

- Energy-Aware Processing:

The processing location can be selected based on energy consumption, allowing computational tasks to be shifted between centralized and distributed units to improve overall energy efficiency.

- Multi-RU Coordination:

Coordinated beamforming and joint resource allocation between neighboring radio units can be implemented to improve spectrum utilization and reduce inter-cell interference.

5.Short Design Outcome

Centralized Unit → Control + Scheduling + Resource Management

↓

Fronthaul → Selected processed information

↓

Distributed Unit/Radio Unit → Baseband + Modulation + Beamforming + RF

↓

Antenna → Wireless Transmission

↓

User

The proposed functional split model balances centralized intelligence with distributed radio processing, thereby reducing fronthaul overhead, lowering latency, improving resource utilization, and providing a scalable architecture for 5G/6G wireless communication systems.

6. MATLAB Design and Simulation

```
clc;
clear;
close all;

%% Functional Split Model
% Number of Distributed Radio Units
N = 8;

% Traffic generated by each RU (Mbps)
traffic = [20 25 30 35 40 45 50 55];

% Percentage of processing performed locally at RU
RU_processing = 0.60;

% Percentage forwarded to CU
CU_processing = 0.40;

%% Fully Centralized Architecture

centralized_processing = traffic;
centralized_fronthaul = traffic;

%% Functional Split Architecture

distributed_processing = RU_processing .* traffic;
split_fronthaul = CU_processing .* traffic;

%% Total Processing and Fronthaul

total_centralized_processing = sum(centralized_processing);
total_distributed_processing = sum(distributed_processing);

total_centralized_fronthaul = sum(centralized_fronthaul);
total_split_fronthaul = sum(split_fronthaul);

%% Fronthaul Reduction

fronthaul_reduction = ...
    ((total_centralized_fronthaul - total_split_fronthaul) ...
    / total_centralized_fronthaul) * 100;

%% Display Results

fprintf('\n===== \n');
fprintf(' FUNCTIONAL SPLIT PERFORMANCE ANALYSIS \n');
fprintf('===== \n');

fprintf('\nNumber of Radio Units      : %d \n', N);

fprintf('Total Traffic                    : %.2f Mbps \n', ...
    sum(traffic));

fprintf('\nFully Centralized: \n');
fprintf('Processing Load                 : %.2f Mbps \n', ...
    total_centralized_processing);

fprintf('Fronthaul Requirement           : %.2f Mbps \n', ...
    total_centralized_fronthaul);
```

```

fprintf('\nFunctional Split:\n');
fprintf('RU Local Processing      : %.2f Mbps\n', ...
        total_distributed_processing);

fprintf('CU Processing              : %.2f Mbps\n', ...
        CU_processing * sum(traffic));

fprintf('Fronthaul Requirement      : %.2f Mbps\n', ...
        total_split_fronthaul);

fprintf('\nFronthaul Reduction    : %.2f %%\n', ...
        fronthaul_reduction);

%% Graph 1: Processing Distribution

figure;

bar(traffic);

hold on;

bar(distributed_processing);

xlabel('Distributed Radio Unit');
ylabel('Processing Load (Mbps)');

title('Processing Distribution in Functional Split');

legend('Total RU Traffic', ...
        'Local RU Processing');

grid on;

%% Graph 2: Fronthaul Comparison

figure;

bar([centralized_fronthaul' split_fronthaul']);

xlabel('Distributed Radio Unit');
ylabel('Fronthaul Data Rate (Mbps)');

title('Fronthaul Requirement Comparison');

legend('Fully Centralized', ...
        'Functional Split');

grid on;

%% Graph 3: Overall Comparison

figure;

comparison = [total_centralized_fronthaul ...
              total_split_fronthaul];

bar(comparison);

set(gca, 'XTickLabel', ...
        {'Fully Centralized', 'Functional Split'});

```

```

ylabel('Total Fronthaul Requirement (Mbps)');

title('Overall Fronthaul Efficiency');

grid on;

```

Design Outcome

